MedAgent Clinical Decision Support

Case ID: case_demo_pneumonia | **Status:** APPROVE | **Physician:** Dr. Sarah Chen, MD | **Date:** 2026-10-12 18:23 UTC

CRITICAL DISCLAIMER: Decision-support draft generated by AI agents. It is not a diagnosis and must be reviewed, corrected and approved by a licensed physician before any clinical use.

Patient Intake & Presentation

Patient: Age: 64, Sex: male
Notes: 64yo male with 4 days of fever, purulent cough, right pleuritic chest pain and dyspnea.

Clinical Summary / Impression

Patient evaluation demonstrates clinical features consistent with Community-Acquired Pneumonia. Diagnostic synthesis is supported by 5 key multimodal findings. Differential considerations include Sepsis / Severe Systemic Infection, Pleural Effusion. Diagnostic workup and clinical monitoring are indicated to confirm etiology and guide definitive management.

Ranked Differential Diagnoses

#	Condition	Likelihood	Key Supporting Findings	Citations
1	Community-Acquired Pneumonia	High	fever, cough, pleuritic, dyspnea, Pneumonia (p=62.8%)	S2, S3
2	Sepsis / Severe Systemic Infection	Moderate	fever, Leukocytosis with Neutrophilia, Raised Inflammatory Markers	S1, S4
3	Pleural Effusion	Moderate	pleuritic, dyspnea, Raised Inflammatory Markers	S8
4	COPD Acute Exacerbation	Moderate	dyspnea, Leukocytosis with Neutrophilia	S6, S7
5	Normocytic Anemia of Chronic Disease	Moderate	Raised Inflammatory Markers	S_Normoc_1, S_Nor

Radiographic Findings (Chest X-Ray)

Pathology	Probability	Location	Indicative Severity	Confidence
Enlarged Cardiomediastinum	62.9%	cardiac / mediastinal silhouette	Moderate	Moderate
Pneumonia	62.8%	bilateral lower zone	Moderate	Moderate
Emphysema	62.5%	left mid zone	Moderate	Moderate
Fracture	62.5%	bilateral lower zone	Moderate	Moderate
Lung Opacity	59.5%	bilateral mid zone	Moderate	Moderate
Mass	57.8%	bilateral lower zone	Low	Moderate

Laboratory Findings & Identified Patterns

Identified Clinical Patterns: Leukocytosis with Neutrophilia (soon); **Raised Inflammatory Markers** (routine)

Physician Attestation: I, Dr. Dr. Sarah Chen, MD, hereby confirm that I have reviewed, verified, and adjusted this decision-support synthesis. Decision: **APPROVE**.
Reviewer Notes: Clinical differential and radiographic findings verified. Consistent with bacterial CAP.
Signature: _____ **Date:** 2026-10-12